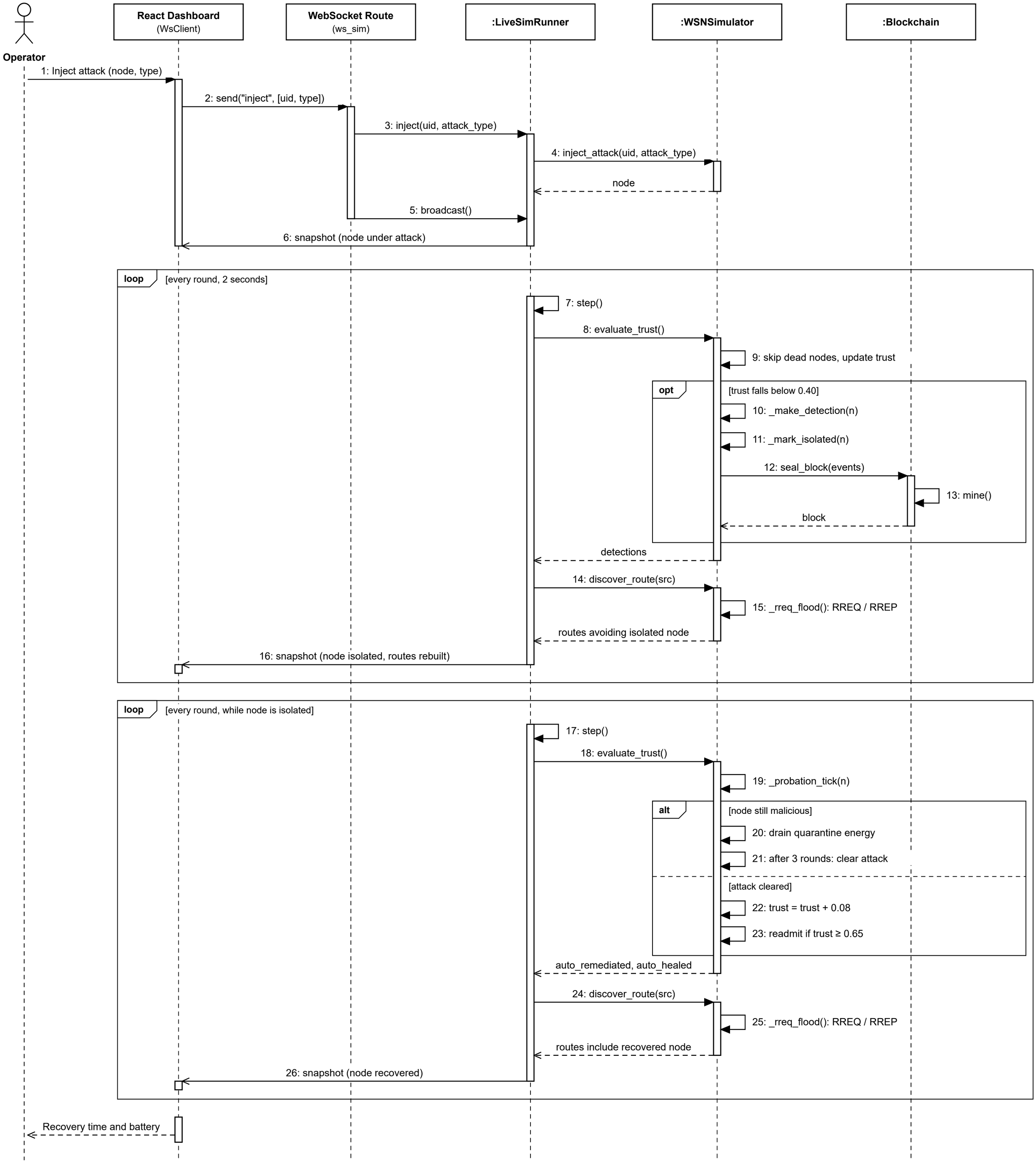

Operator
React Dashboard
(WsClient)
WebSocket Route
(ws_sim)
:LiveSimRunner
:WSNSimulator
:Blockchain
1: Inject attack (node, type)
2: send("inject", [uid, type])
3: inject(uid, attack_type)
4: inject_attack(uid, attack_type)
node
5: broadcast()
6: snapshot (node under attack)
loop
[every round, 2 seconds]
7: step()
8: evaluate_trust()
9: skip dead nodes, update trust
opt
[trust falls below 0.40]
10: _make_detection(n)
11: _mark_isolated(n)
12: seal_block(events)
13: mine()
block
detections
14: discover_route(src)
15: _rreq_flood(): RREQ / RREP
routes avoiding isolated node
16: snapshot (node isolated, routes rebuilt)
loop
[every round, while node is isolated]
17: step()
18: evaluate_trust()
19: _probation_tick(n)
alt
[node still malicious]
20: drain quarantine energy
21: after 3 rounds: clear attack
[attack cleared]
22: trust = trust + 0.08
23: readmit if trust ≥ 0.65
auto_remediated, auto_healed
24: discover_route(src)
25: _rreq_flood(): RREQ / RREP
routes include recovered node
26: snapshot (node recovered)
Recovery time and battery